Chemical and Ionic Equilibrium

A Complete, Self-Contained Master Tutorial with Reference Data, Worked Examples, and Original NEET UG Practice Assessments

1. Learning Objectives & Chapter Overview

Chemical equilibria govern vital biological and industrial systems. For example, the reversible binding of $\mathrm{O_2}$ molecules with the protein haemoglobin enables oxygen transport from our lungs to our muscles, while a competing equilibrium involving CO and haemoglobin explains carbon monoxide toxicity. After studying this tutorial, you will be able to:

- Identify the dynamic nature of equilibrium in both physical phase transformations and chemical reactions.
- Formulate the Law of Chemical Equilibrium, write expressions for K_c and K_p, and establish the quantitative relationship $K_p = K_c(RT)^{\Delta n}$.
- Predict the extent of a reaction from the magnitude of K, determine the direction of net reaction using the reaction quotient Q, and link K to standard Gibbs energy change (ΔG°).
- Apply Le Chatelier's principle to predict how changes in concentration, partial pressure, volume, inert gas addition, temperature, and catalysts affect equilibrium state.
- Classify acids and bases using Arrhenius, Brönsted-Lowry, and Lewis concepts, and identify conjugate acid-base pairs.
- Calculate degree of ionization (α), ionization constants (K_a, K_b), ionic product of water (K_w), and pH across strong electrolytes, weak electrolytes, and very dilute acids.
- Analyze the common ion effect, salt hydrolysis across different salt categories, buffer action using the Henderson-Hasselbalch equation, and solubility equilibria (K_{sp}) including selective precipitation and pH-dependent solubility.

2. Equilibrium in Physical Processes

When a liquid evaporates in a closed container, molecules with relatively higher kinetic energy escape the liquid surface into the vapour phase, while vapour molecules strike the liquid surface and condense back into the liquid phase. Eventually, a constant vapour pressure is established because the number of molecules leaving the liquid equals the number returning from the vapour. At this stage, the system has reached a **dynamic equilibrium**:

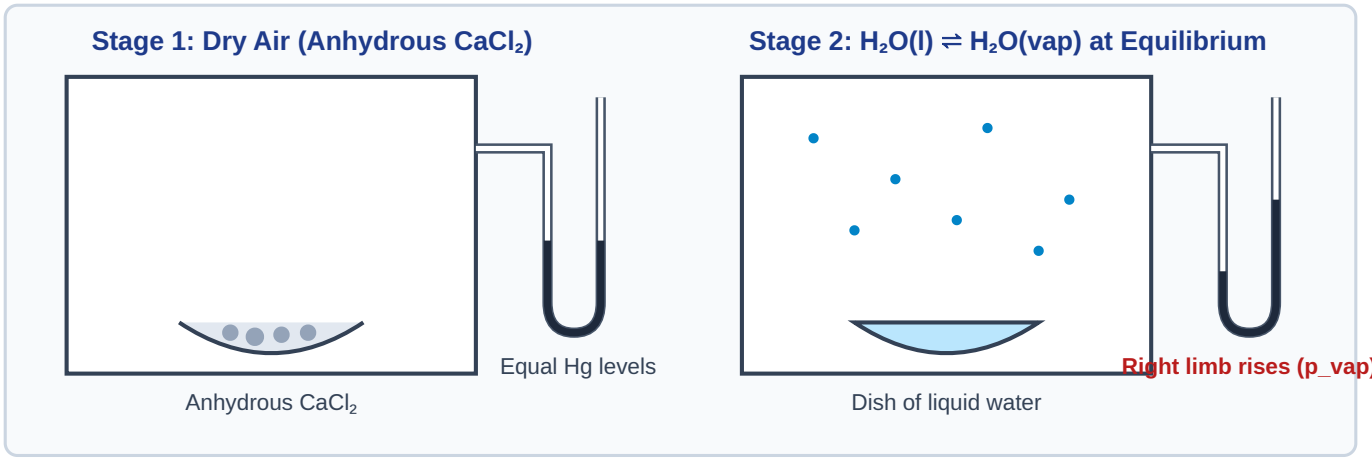
$$\mathrm{H_2O(l)} \rightleftharpoons \mathrm{H_2O(vap)} \quad (\text{Rate of evaporation} = \text{Rate of condensation})$$

The double half-arrows ($\rightleftharpoons$) indicate that both opposing processes occur simultaneously at equal rates. The mixture of species at equilibrium is called an **equilibrium mixture**.

2.1 Phase Transformation Equilibria

- **Solid–Liquid Equilibrium:** Ice and water kept in a perfectly insulated thermos flask at $273\ \mathrm{K}$ and atmospheric pressure ($1.013\ \mathrm{bar}$) coexist with constant mass of both phases. Molecules of liquid water collide with ice and adhere to it at the exact same rate at which ice molecules escape into the liquid phase. For any pure substance at $1\ \mathrm{atm}$ ($1.013\ \mathrm{bar}$) pressure, the unique temperature at which solid and liquid phases coexist at equilibrium is called the **normal melting point** or **normal freezing point**.
- **Liquid–Vapour Equilibrium:** Consider a transparent closed box fitted with a mercury U-tube manometer. Initially, the air inside is dried using a desiccant such as anhydrous calcium chloride ($\mathrm{CaCl_2}$) or phosphorus pentoxide ($\mathrm{P_4O_{10}}$). When the drying agent is removed and a watch glass containing liquid water is quickly placed inside at constant temperature, the mercury level in the right limb of the manometer slowly rises and reaches a constant value while the volume of liquid water decreases. Initially, the rate of evaporation is constant and condensation is negligible. As vapour accumulates, the rate of condensation increases until $\text{rate of evaporation} = \text{rate of condensation}$. The constant pressure exerted by water molecules at equilibrium at a given temperature is the **equilibrium vapour pressure**. Different liquids (e.g., methyl alcohol, acetone, ether, water) have different equilibrium vapour pressures at the same temperature. A liquid with a higher vapour pressure is more *volatile* and has a lower boiling point. In an open watch glass, vapour disperses into the room, condensation is negligible, and equilibrium can never be reached. For any pure liquid at $1.013\ \mathrm{bar}$ pressure, the temperature at which liquid and vapour coexist at equilibrium is the **normal boiling point** ($100\ ^\circ\mathrm{C}$ for water). At high altitudes, atmospheric pressure decreases, so the boiling point decreases.
- **Solid–Vapour Equilibrium (Sublimation):** When solid iodine is placed in a closed vessel, violet vapours appear and deepen in colour until the colour intensity becomes constant. At this point, the rate of sublimation equals the rate of deposition/condensation:
$$\mathrm{I_2(solid)} \rightleftharpoons \mathrm{I_2(vapour)}$$
Other examples include $\mathrm{Camphor(solid)} \rightleftharpoons \mathrm{Camphor(vapour)}$ and $\mathrm{NH_4Cl(solid)} \rightleftharpoons \mathrm{NH_4Cl(vapour)}$.

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Tutorial Diagram 1: Measurement of equilibrium vapour pressure of water at constant temperature using a closed chamber and U-tube mercury manometer.

2.2 Dissolution of Solids and Gases in Liquids

- **Solids in Liquids:** At a given temperature, a solution in which no more solute can be dissolved is called a **saturated solution**. In a saturated solution, a dynamic equilibrium exists between undissolved solid solute and dissolved solute:
$$\text{Sugar(solution)} \rightleftharpoons \text{Sugar(solid)} \quad (\text{Rate of dissolution} = \text{Rate of crystallisation})$$
When radioactive sugar crystals are added to a saturated solution of non-radioactive sugar, radioactivity is soon detected in both the solution and the solid phase, and the ratio of radioactive to non-radioactive molecules in solution increases until it attains a constant value. This confirms the dynamic exchange of molecules across phases at equal rates.
- **Gases in Liquids:** In a sealed soda-water bottle, carbon dioxide gas is in dynamic equilibrium with dissolved carbon dioxide molecules:
$$\mathrm{CO_2(gas)} \rightleftharpoons \mathrm{CO_2(in\ solution)}$$
This equilibrium is governed by **Henry's Law**, which states that the mass of a gas dissolved in a given mass of a solvent at any temperature is proportional to the pressure of the gas above the solvent. Solubility of a gas in a liquid decreases with an increase in temperature. When the bottle is opened, the partial pressure of $\mathrm{CO_2}$ above the liquid drops sharply to atmospheric partial pressure, causing dissolved $\mathrm{CO_2}$ to fizz out to establish a new equilibrium state.

Physical Equilibrium Process	Characteristic Constant Quantity at a Given Temperature
$\text{Liquid} \rightleftharpoons \text{Vapour}$, e.g., $\mathrm{H_2O(l)} \rightleftharpoons \mathrm{H_2O(g)}$	Vapour pressure ($p_{\mathrm{H_2O}}$) is constant at a given temperature.
$\text{Solid} \rightleftharpoons \text{Liquid}$, e.g., $\mathrm{H_2O(s)} \rightleftharpoons \mathrm{H_2O(l)}$	Melting point is fixed at constant pressure ($273\ \mathrm{K}$ at $1.013\ \mathrm{bar}$).
$\text{Solute(s)} \rightleftharpoons \text{Solute(solution)}$, e.g., $\text{Sugar(s)} \rightleftharpoons \text{Sugar(solution)}$	Concentration of solute in the saturated solution (solubility) is constant at a given temperature.
$\text{Gas(g)} \rightleftharpoons \text{Gas(aq)}$, e.g., $\mathrm{CO_2(g)} \rightleftharpoons \mathrm{CO_2(aq)}$	Ratio $\frac{[\text{gas(aq)}]}{[\text{gas(g)}]}$ (or $\frac{[\mathrm{CO_2(aq)}]}{[\mathrm{CO_2(g)}]}$) is constant at a given temperature.

2.3 General Characteristics of Physical Equilibria

1. Equilibrium is possible only in a **closed system** at a given temperature.
2. Both opposing processes occur simultaneously at the exact same rate, creating a **dynamic** yet stable state.
3. All measurable macroscopic properties of the system (pressure, concentration, colour, density) remain constant.
4. Attainment of physical equilibrium is characterized by a constant value of one of its parameters at a given temperature.
5. The magnitude of these characteristic quantities at any stage indicates the extent to which the physical process has proceeded before reaching equilibrium.

3. Equilibrium in Chemical Processes — Dynamic Nature

Reversible chemical reactions occur in both forward and backward directions. Consider a general reversible reaction in a closed vessel:
$$\mathrm{A + B \rightleftharpoons C + D}$$
Initially, if we start only with reactants A and B, the forward reaction rate is high while the reverse reaction rate is zero. With time, A and B deplete and products C and D accumulate. Consequently, the rate of the forward reaction decreases and the rate of the reverse reaction increases until both rates become equal. At this stage, the system attains **dynamic chemical equilibrium**, and the concentrations of A, B, C, and D remain constant with time.

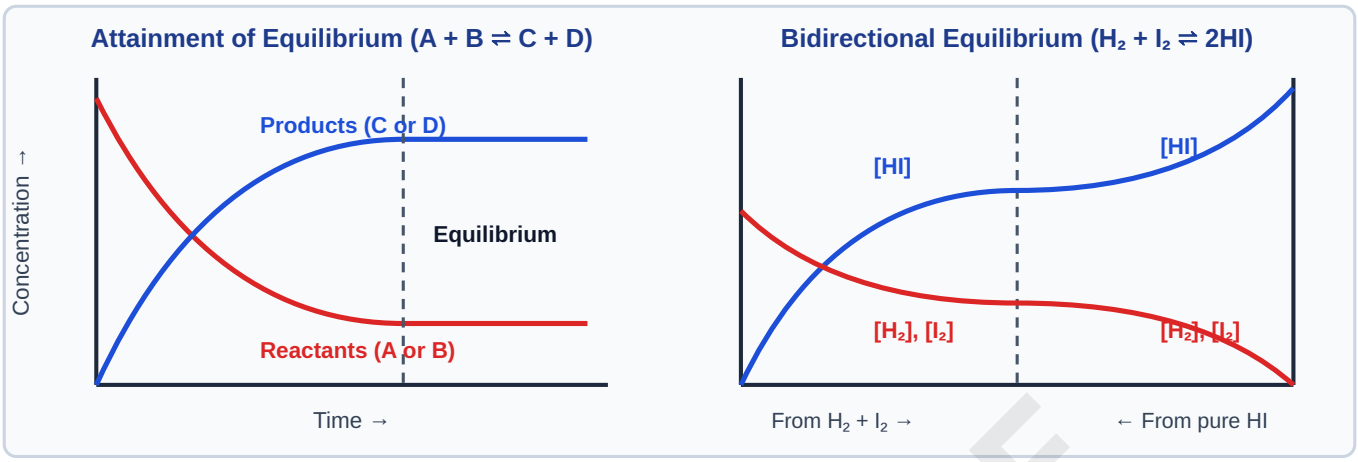
Experimental Proof of Dynamic Nature & Bidirectional Attainment

- **Isotope Scrambling in Haber's Process:** In the synthesis of ammonia, $\mathrm{N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)}$, if one equilibrium mixture prepared from $\mathrm{N_2}$ and $\mathrm{H_2}$ (containing $\mathrm{N_2, H_2, NH_3}$) and a second equilibrium mixture prepared under identical conditions using deuterium $\mathrm{D_2}$ (containing $\mathrm{N_2, D_2, ND_3}$) are mixed together, the total ammonia concentration remains unchanged. However, mass spectrometry reveals scrambled species: $\mathrm{NH_3, NH_2D, NHD_2, ND_3}$ as well as $\mathrm{H_2, HD, D_2}$. This proves that forward and reverse reactions continue at equal rates even after equilibrium is reached.
- **Attainment from Either Direction:** In the reaction $\mathrm{H_2(g) + I_2(g) \rightleftharpoons 2HI(g)}$, provided the total number of H and I atoms in a given volume at a given temperature is the same, the exact same equilibrium composition is obtained whether we start with pure reactants (

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$H_2 + I_2$) or pure product (HI).

- **Two-Cylinder Transfer Analogy:** If coloured water ($KMnO_4$ solution) is repeatedly transferred back and forth between two $100\,mL$ measuring cylinders using two glass tubes, the liquid levels in both cylinders eventually become constant even though liquid transfer continues in both directions. If tubes of different diameters are used, the final equilibrium levels in the two cylinders are different.



Tutorial Diagram 2: (Left) Evolution of reactant and product concentrations to equilibrium. (Right) Attainment of the same equilibrium state from either pure reactants or pure product.

4. Law of Chemical Equilibrium & Equilibrium Constant

In 1864, Norwegian chemists Cato Maximillian Guldberg and Peter Waage proposed the **Law of Mass Action** (also known as the **Law of Chemical Equilibrium**): *At a given temperature, the product of equilibrium concentrations of the reaction products raised to their respective stoichiometric coefficients in the balanced chemical equation divided by the product of equilibrium concentrations of the reactants raised to their individual stoichiometric coefficients has a constant value.*

For a general balanced reversible reaction:

$$aA + bB \rightleftharpoons cC + dD$$

$$K_c = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

Here, $[A]$, $[B]$, $[C]$, and $[D]$ represent equilibrium molar concentrations ($mol\,L^{-1}$). For example, for $4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(g)$, the equilibrium constant is $K_c = \frac{[NO]^4[H_2O]^6}{[NH_3]^4[O_2]^5}$.

4.1 Experimental Verification ($H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ at $731\,K$)

When gaseous H_2 and I_2 react in a sealed vessel at $731\,K$ (experiments 1 to 4 starting from $H_2 + I_2$) or when HI decomposes (experiments 5 and 6 starting from pure HI), the simple ratio $\frac{[HI]_{eq}}{[H_2]_{eq}[I_2]_{eq}}$ varies widely (from 790 to 1970), whereas $\frac{[HI]^2_{eq}}{[H_2]_{eq}[I_2]_{eq}}$ remains essentially constant around 46.4–47.6:

Exp.	Initial $[H_2]$ (M)	Initial $[I_2]$ (M)	Initial $[HI]$ (M)	Equil. $[H_2]$ (M)	Equil. $[I_2]$ (M)	Equil. $[HI]$ (M)	$\frac{[HI]^2_{eq}}{[H_2]_{eq}[I_2]_{eq}}$
1	2.4×10^{-2}	1.38×10^{-2}	0	1.14×10^{-2}	0.12×10^{-2}	2.52×10^{-2}	46.4
2	2.4×10^{-2}	1.68×10^{-2}	0	0.92×10^{-2}	0.20×10^{-2}	2.96×10^{-2}	47.6
3	2.44×10^{-2}	1.98×10^{-2}	0	0.77×10^{-2}	0.31×10^{-2}	3.34×10^{-2}	46.7
4	2.46×10^{-2}	1.76×10^{-2}	0	0.92×10^{-2}	0.22×10^{-2}	3.08×10^{-2}	46.9
5	0	0	3.04×10^{-2}	0.345×10^{-2}	0.345×10^{-2}	2.35×10^{-2}	46.4
6	0	0	7.58×10^{-2}	0.86×10^{-2}	0.86×10^{-2}	5.86×10^{-2}	46.4

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4.2 Mathematical Rules for Manipulating Equilibrium Constants

Operation on Balanced Chemical Equation	New Chemical Equation	New Equilibrium Constant
Original forward reaction	$aA + bB \rightleftharpoons cC + dD$	K_c
Reversing the reaction	$cC + dD \rightleftharpoons aA + bB$	$K_c' = \dfrac{1}{K_c}$
Multiplying stoichiometric coefficients by n	$naA + nbB \rightleftharpoons ncC + ndD$	$K_c'' = (K_c)^n$
Adding two or more reactions ($R_{\text{net}} = R_1 + R_2$)	Sum of individual equations	$K_{\text{NET}} = K_1 \times K_2$

4.3 Homogeneous & Heterogeneous Equilibria and Relationship Between K_p and K_c

In a **homogeneous system**, all reactants and products are in the same phase—for example, $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ in the gas phase, or $CH_3COOC_2H_5(aq) + H_2O(l) \rightleftharpoons CH_3COOH(aq) + C_2H_5OH(aq)$ and $Fe^{3+}(aq) + SCN^-(aq) \rightleftharpoons [Fe(SCN)]^{2+}(aq)$ in the solution phase.

For gaseous reactions, using the ideal gas law $pV = nRT \implies p = \dfrac{n}{V}RT = cRT = [\text{gas}]RT$, where c is in mol L^{-1}, p is in bar, and $R = 0.0831\ \text{bar L mol}^{-1}\ \text{K}^{-1}$ (note that $1\ \text{bar} = 10^5\ \text{Pa} = 10^5\ \text{N m}^{-2}$), the partial pressure equilibrium constant K_p for $aA(g) + bB(g) \rightleftharpoons cC(g) + dD(g)$ is:

$$K_p = \frac{(p_C)^c (p_D)^d}{(p_A)^a (p_B)^b} = \frac{[C]^c [D]^d}{[A]^a [B]^b} (RT)^{(c+d)-(a+b)} \implies K_p = K_c (RT)^{\Delta n}$$

where $\Delta n = (\text{number of moles of gaseous products}) - (\text{number of moles of gaseous reactants})$ in the balanced chemical equation. For $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$, $\Delta n = 2 - 2 = 0$, so $K_p = K_c$. For $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, $\Delta n = 2 - 4 = -2$, so $K_p = K_c (RT)^{-2}$.

Selected Gaseous Reaction	Temperature (K)	K_p Value
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$	298 K / 400 K / 500 K	6.8×10^5 / 41 / 3.6×10^{-2}
$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$	298 K / 500 K / 700 K	4.0×10^{24} / 2.5×10^{10} / 3.0×10^4
$N_2O_4(g) \rightleftharpoons 2NO_2(g)$	298 K / 400 K / 500 K	0.98 / 47.9 / 1700

In a **heterogeneous equilibrium**, reactants and products exist in more than one phase. Because the molar concentration (density divided by molar mass) of a **pure solid** or a **pure liquid** is constant at a given temperature and independent of the amount present, pure solids and pure liquids are omitted from K_c and K_p expressions (though they must be physically present in the container for the equilibrium to exist):

- $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$: $K_c = [CO_2(g)]$, $K_p = p_{CO_2}$ (At 1100 K, $p_{CO_2} = 2.0 \times 10^5\ \text{Pa} = 2.00\ \text{bar} \implies K_p = 2.00$).
- $Ni(s) + 4CO(g) \rightleftharpoons Ni(CO)_4(g)$: $K_c = \dfrac{[Ni(CO)_4]}{[CO]^4}$
- $Ag_2O(s) + 2HNO_3(aq) \rightleftharpoons 2AgNO_3(aq) + H_2O(l)$: $K_c = \dfrac{[AgNO_3]^2}{[HNO_3]^2}$

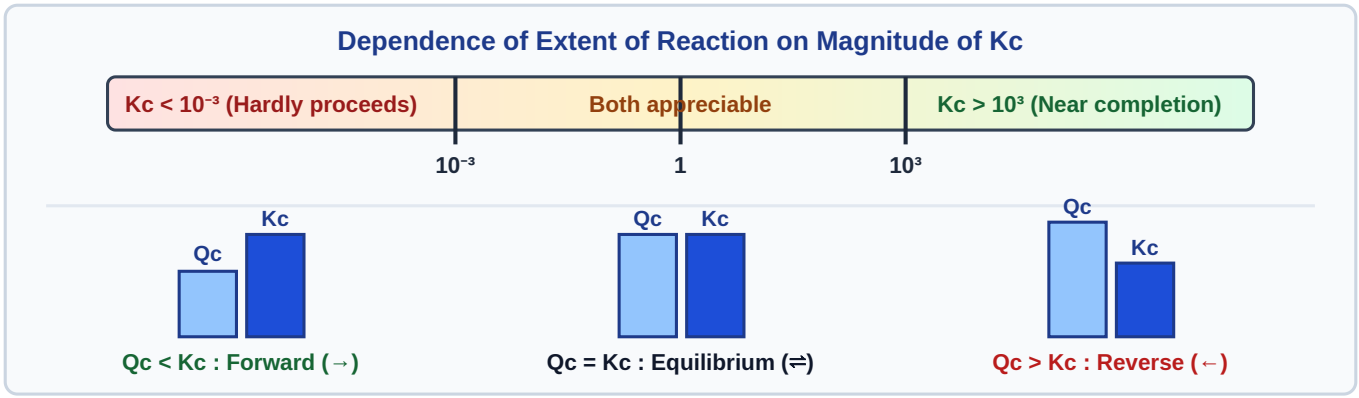
Units and Dimensionless Form of K: When concentrations are substituted in mol L^{-1} and pressures in bar (or atm), K_c has units of $(\text{mol L}^{-1})^{\Delta n}$ and K_p has units of $(\text{bar})^{\Delta n}$. However, when activities are defined relative to standard states—$p^\circ = 1\ \text{bar}$ for gases and $c^\circ = 1\ \text{M}$ for solutes—both K_p and K_c are strictly **dimensionless quantities** (though they have different numerical values when $\Delta n \neq 0$ due to different standard states).

5. Applications of Equilibrium Constants, Gibbs Energy & Le Chatelier's Principle

5.1 Predicting the Extent and Direction of a Reaction

- **Extent of Reaction (from magnitude of K_c):**
  - If $K_c > 10^3$, products predominate strongly over reactants and the reaction proceeds nearly to completion. Examples: $2H_2(g) + O_2(g) \rightleftharpoons 2H_2O(g)$ at 500 K ($K_c = 2.4 \times 10^{47}$); $H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$ at 300 K ($K_c = 4.0 \times 10^{31}$); $H_2(g) + Br_2(g) \rightleftharpoons 2HBr(g)$ at 300 K ($K_c = 5.4 \times 10^{18}$).
  - If $K_c < 10^{-3}$, reactants predominate strongly over products and the forward reaction proceeds rarely. Examples: decomposition of H_2O at 500 K ($K_c = 4.1 \times 10^{-48}$); $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$ at 298 K ($K_c = 4.8 \times 10^{-31}$).
  - If $10^{-3} \leq K_c \leq 10^3$, appreciable concentrations of both reactants and products are present at equilibrium. Examples: $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ at 700 K ($K_c = 57.0$); $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ at 25 °C ($K_c = 4.64 \times 10^{-3}$).
- **Direction of Reaction (using Reaction Quotient Q):** At any arbitrary time t, the reaction quotient $Q_c = \dfrac{[C]_t^c [D]_t^d}{[A]_t^a [B]_t^b}$ (or Q_p with partial pressures) is compared with K_c:
  - If $Q_c < K_c$: Net reaction proceeds from **left to right (forward direction)** until $Q_c = K_c$.
  - If $Q_c > K_c$: Net reaction proceeds from **right to left (reverse direction)** until $Q_c = K_c$.
  - If $Q_c = K_c$: The reaction mixture is already at **equilibrium**; no net reaction occurs.

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Tutorial Diagram 3: Predicting reaction extent from K_c and predicting net reaction direction by comparing Q_c with K_c.

5.2 Relationship Between Equilibrium Constant K, Reaction Quotient Q, and Gibbs Energy G

The equilibrium constant does not depend on the rate of reaction, but is directly governed by reaction thermodynamics (ΔG):

- If $\Delta G < 0$, the reaction is spontaneous and proceeds in the forward direction.
- If $\Delta G > 0$, the forward reaction is non-spontaneous and the reverse reaction proceeds spontaneously.
- If $\Delta G = 0$, the reaction has achieved equilibrium.

At any stage of the reaction, $\Delta G = \Delta G^\circ + RT \ln Q$. At equilibrium, $\Delta G = 0$ and $Q = K$, yielding:

$$\Delta G^\circ = -RT \ln K = -2.303\, RT \log K \quad \Longleftrightarrow \quad K = e^{-\Delta G^\circ / RT}$$

Thus, if $\Delta G^\circ < 0$, then $K > 1$ (products predominate at equilibrium). If $\Delta G^\circ > 0$, then $K < 1$ (reactants predominate at equilibrium).

5.3 Le Chatelier's Principle and Factors Affecting Equilibria

Le Chatelier's Principle states that *a change in any of the factors that determine the equilibrium conditions of a system will cause the system to change in such a manner so as to reduce or to counteract the effect of the change.*

Applied Stress / Change	Effect on Equilibrium Position & K	Illustrative Examples from Syllabus
1. Change in Concentration	Adding a reactant (or removing a product) makes $Q_c < K_c$, shifting equilibrium **forward**. Adding a product (or removing a reactant) makes $Q_c > K_c$, shifting equilibrium **backward**. K_c remains unchanged.	• Liquefying and removing $\mathrm{NH_3}$ in Haber's process, or venting $\mathrm{CO_2}$ from the kiln during $\mathrm{CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)}$, drives the forward reaction. • In $\mathrm{Fe^{3+}(aq, yellow) + SCN^-(aq, colourless) \rightleftharpoons [Fe(SCN)]^{2+}(aq, deep\ red)}$, adding KSCN deepens the red colour (shifts right). Adding oxalic acid ($\mathrm{H_2C_2O_4}$) removes $\mathrm{Fe^{3+}}$ as stable $\mathrm{[Fe(C_2O_4)_3]^{3-}}$, or adding $\mathrm{HgCl_2}$ removes $\mathrm{SCN^-}$ as stable $\mathrm{[Hg(SCN)_4]^{2-}}$; both shift the equilibrium left and fade the red colour.
2. Change in Pressure / Volume	Increasing pressure (by decreasing volume) shifts a gaseous equilibrium toward the side with **fewer moles of gas**. Decreasing pressure shifts toward **more moles of gas**. If $\Delta n = 0$, pressure/volume change has no effect on composition. K_c is unchanged.	• In methanation, $\mathrm{CO(g) + 3H_2(g) \rightleftharpoons CH_4(g) + H_2O(g)}$ ($4\,\mathrm{mol\ gas} \rightarrow 2\,\mathrm{mol\ gas}$), halving the volume doubles all concentrations, making $Q_c = \frac{(2[\mathrm{CH_4}])(2[\mathrm{H_2O}])}{(2[\mathrm{CO}])(2[\mathrm{H_2}])^3} = \frac{1}{4}K_c < K_c$, shifting equilibrium forward. • In $\mathrm{C(s) + CO_2(g) \rightleftharpoons 2CO(g)}$, increasing pressure shifts equilibrium backward.
3. Addition of Inert Gas	At **constant volume**, adding an inert gas (like argon) does not alter partial pressures or molar concentrations of reacting species, so the equilibrium remains completely undisturbed.	Adding $\mathrm{Ar(g)}$ at constant V to any equilibrium mixture causes no shift.
4. Change in Temperature	**Alters the numerical value of K_c!** • For an **exothermic reaction ($\Delta H < 0$)**, increasing T decreases K_c (shifts backward); cooling increases K_c (shifts forward). • For an **endothermic reaction ($\Delta H > 0$)**, increasing T increases K_c (shifts forward); cooling decreases K_c (shifts backward).	• Exothermic dimerisation: $\mathrm{2NO_2(g, brown) \rightleftharpoons N_2O_4(g, colourless)}$, $\Delta H = -57.2\,\mathrm{kJ\,mol^{-1}}$. In a freezing bath ($270\,\mathrm{K}$), forward reaction is favoured and brown colour fades; in hot water ($363\,\mathrm{K}$), reverse reaction is favoured and brown colour intensifies. • Endothermic complexation: $\mathrm{[Co(H_2O)_6]^{2+}(aq, pink) + 4Cl^-(aq) \rightleftharpoons [CoCl_4]^{2-}(aq, blue) + 6H_2O(l)}$. At room temperature it is blue; cooling in a freezing mixture shifts it left, turning the solution pink.
5. Effect of a Catalyst	Lowers the activation energy of both forward and reverse reactions by the exact same amount. Speeds up attainment of equilibrium, but **does not change K_c or equilibrium composition**.	• In Haber's process ($\mathrm{N_2 + 3H_2 \rightleftharpoons 2NH_3}$, $\Delta H = -92.38\,\mathrm{kJ\,mol^{-1}}$), an iron catalyst allows satisfactory rate at optimum conditions ($\sim 500\,^\circ\mathrm{C}, 200\,\mathrm{atm}$). • In the Contact Process ($\mathrm{2SO_2 + O_2 \rightleftharpoons 2SO_3}$, $K_c = 1.7 \times 10^{26}$), Pt or $\mathrm{V_2O_5}$ overcomes the slow oxidation rate. (If K is exceedingly small, a catalyst is of no help.)

6. Ionic Equilibrium: Concepts of Acids, Bases, and Salts

Michael Faraday classified substances into **electrolytes** (which conduct electricity in aqueous solution due to ions) and **non-electrolytes** (such as sugar, which do not conduct). Electrolytes are further divided into **strong electrolytes** (almost 100% ionized in water, e.g., $\mathrm{NaCl, HCl, NaOH}$)

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and **weak electrolytes** ($< 5\%$ ionized, e.g., $\mathrm{CH_3COOH}, \mathrm{NH_3}$, where an **ionic equilibrium** is established between ions and unionized molecules). Water has a very high dielectric constant of 80, which reduces electrostatic attractions between oppositely charged ions (such as $\mathrm{Na^+}$ and $\mathrm{Cl^-}$) by a factor of 80 and stabilizes the separated ions via hydration.

6.1 Three Classical Theories of Acids and Bases

1. **Arrhenius Concept:** Acids dissociate in water to give hydrogen ions $\mathrm{H^+(aq)}$ (or hydronium ions $\mathrm{H_3O^+(aq)}$), while bases dissociate in water to give hydroxyl ions $\mathrm{OH^-(aq)}$. Because a bare proton ($\mathrm{H^+}$, radius $\sim 10^{-15}\,\mathrm{m}$) has an intense electric field, it bonds to a lone pair on a water oxygen atom to form a trigonal pyramidal hydronium ion $\mathrm{H_3O^+}$, which is further hydrated in solution to species like $\mathrm{H_5O_2^+}, \mathrm{H_7O_3^+}$, and $\mathrm{H_9O_4^+}$. Similarly, $\mathrm{OH^-}$ is hydrated to $\mathrm{H_3O_2^-}, \mathrm{H_5O_3^-}, \mathrm{H_7O_4^-}$. *Limitation:* Restricted to aqueous solutions and cannot explain the basicity of substances lacking an $-\mathrm{OH}$ group (such as $\mathrm{NH_3}$).
2. **Brönsted–Lowry Concept (1923):** An **acid is a proton ($\mathrm{H^+}$) donor** and a **base is a proton ($\mathrm{H^+}$) acceptor**. An acid-base pair that differs only by one proton ($\mathrm{H^+}$) is called a **conjugate acid-base pair**:
$$\underbrace{\mathrm{NH_3(aq)}}_{\text{Base}} + \underbrace{\mathrm{H_2O(l)}}_{\text{Acid}} \rightleftharpoons \underbrace{\mathrm{NH_4^+(aq)}}_{\text{Conjugate acid}} + \underbrace{\mathrm{OH^-(aq)}}_{\text{Conjugate base}}$$
$$\underbrace{\mathrm{HCl(aq)}}_{\text{Acid}} + \underbrace{\mathrm{H_2O(l)}}_{\text{Base}} \rightleftharpoons \underbrace{\mathrm{H_3O^+(aq)}}_{\text{Conjugate acid}} + \underbrace{\mathrm{Cl^-(aq)}}_{\text{Conjugate base}}$$
   - $\mathrm{Conjugate\ Acid = Base + H^+}; \quad \mathrm{Conjugate\ Base = Acid - H^+}$.
   - **Reciprocal Strength Rule:** A strong Brönsted acid (e.g., $\mathrm{HClO_4, HCl, HBr, HI, HNO_3, H_2SO_4}$) has a very weak conjugate base ($\mathrm{ClO_4^-, Cl^-, Br^-, I^-, NO_3^-, HSO_4^-}$), which is a much weaker base than $\mathrm{H_2O}$. Conversely, a weak acid or very weak proton donor has a strong conjugate base (e.g., $\mathrm{NH_2^-}, \mathrm{O^{2-}}, \mathrm{H^-}$ are much stronger bases than $\mathrm{H_2O}$). Every proton-transfer equilibrium favours the formation of the *weaker acid and weaker base*.
   - **Amphoteric Species:** Species like $\mathrm{H_2O}, \mathrm{HCO_3^-}, \mathrm{HSO_4^-}$, and $\mathrm{NH_3}$ can act as both Brönsted acids and Brönsted bases.
3. **Lewis Concept (G.N. Lewis, 1923):** A **Lewis acid is an electron-pair acceptor** and a **Lewis base is an electron-pair donor**.
$$\underbrace{\mathrm{BF_3}}_{\text{Lewis acid}} + \underbrace{:\mathrm{NH_3}}_{\text{Lewis base}} \longrightarrow \mathrm{BF_3 \leftarrow : NH_3}$$
   - *Common Lewis Acids:* Electron-deficient molecules ($\mathrm{BF_3}, \mathrm{BCl_3}, \mathrm{AlCl_3}$) and cations ($\mathrm{H^+}, \mathrm{Co^{3+}}, \mathrm{Mg^{2+}}, \mathrm{Ag^+}$). Note that $\mathrm{NH_4^+}$ has no vacant orbital to accept a lone pair, so it is a Brönsted acid but *not* a Lewis acid.
   - *Common Lewis Bases:* Neutral molecules with lone pairs ($\mathrm{H_2O}, \mathrm{NH_3}, \mathrm{RNH_2}$) and anions ($\mathrm{OH^-}, \mathrm{F^-}, \mathrm{CN^-}, \mathrm{Cl^-}$).

6.2 Trends in Acid Strength ($\mathrm{H–A}$ Bond Strength vs. Polarity)

The acid strength of HA depends on two factors: the **strength** of the $\mathrm{H–A}$ bond and the **polarity** of the $\mathrm{H–A}$ bond:

- **Down a Group in the Periodic Table: $\mathrm{H–A}$ bond strength** is the deciding factor. As atomic size of A increases down a group, $\mathrm{H–A}$ bond strength decreases, cleavage becomes easier, and acid strength increases:
$$\text{Group 17: } \mathrm{HF} \ll \mathrm{HCl} \ll \mathrm{HBr} \ll \mathrm{HI} \qquad \text{Group 16: } \mathrm{H_2O} < \mathrm{H_2S}$$
- **Across a Period (Same Row) in the Periodic Table: $\mathrm{H–A}$ bond polarity (electronegativity of A)** is the deciding factor. As electronegativity of A increases across a row, bond polarity and acidity increase:
$$\text{Period 2: } \mathrm{CH_4} < \mathrm{NH_3} < \mathrm{H_2O} < \mathrm{HF}$$

7. Ionization of Water, the pH Scale, and Weak Electrolytes

7.1 Self-Ionization of Water and Ionic Product (K_w)

Pure water undergoes auto-protolysis where one water molecule acts as an acid and another as a base:
$$\mathrm{H_2O(l) + H_2O(l) \rightleftharpoons H_3O^+(aq) + OH^-(aq)}$$
Incorporating the constant molarity of pure liquid water ($[\mathrm{H_2O}] = \dfrac{1000\,\mathrm{g\,L^{-1}}}{18.0\,\mathrm{g\,mol^{-1}}} = 55.55\,\mathrm{M}$) into the equilibrium constant gives the **ionic product of water (K_w)**:

$$K_w = [\mathrm{H_3O^+}][\mathrm{OH^-}] = [\mathrm{H^+}][\mathrm{OH^-}] = 1.0 \times 10^{-14}\,\mathrm{M^2} \quad (\text{at } 298\,\mathrm{K})$$

In pure water at $298\,\mathrm{K}$, $[\mathrm{H^+}] = [\mathrm{OH^-}] = 1.0 \times 10^{-7}\,\mathrm{M}$, so the ratio of dissociated to undissociated water is $\dfrac{10^{-7}}{55.55} = 1.8 \times 10^{-9}$. Because auto-ionization of water is endothermic, K_w increases with temperature (e.g., at $310\,\mathrm{K}$, $K_w = 2.7 \times 10^{-14}\,\mathrm{M^2}$).

7.2 The pH Scale

The pH of a solution is defined as the negative base-10 logarithm of the hydrogen ion activity ($a_{\mathrm{H^+}} = [\mathrm{H^+}]/\mathrm{mol\,L^{-1}}$ in dilute solutions $< 0.01\,\mathrm{M}$):

$$\mathrm{pH} = -\log a_{\mathrm{H^+}} \approx -\log[\mathrm{H^+}], \qquad \mathrm{pOH} = -\log[\mathrm{OH^-}], \qquad \mathrm{p}K_w = \mathrm{pH} + \mathrm{pOH} = 14 \quad (\text{at } 298\,\mathrm{K})$$

- At $298\,\mathrm{K}$: **Acidic solution:** $[\mathrm{H_3O^+}] > [\mathrm{OH^-}] \implies \mathrm{pH} < 7$; **Neutral solution:** $[\mathrm{H_3O^+}] = [\mathrm{OH^-}] \implies \mathrm{pH} = 7$; **Basic solution:** $[\mathrm{H_3O^+}] < [\mathrm{OH^-}] \implies \mathrm{pH} > 7$.
- Because the pH scale is logarithmic, decreasing pH by 1 unit corresponds to a 10-fold increase in $[\mathrm{H^+}]$, and a change of 2 units corresponds to a 100-fold change in $[\mathrm{H^+}]$.

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- **Acid-Base Indicators:** Water-soluble weak organic acids (like phenolphthalein and bromothymol blue) exhibit different colours in their unionized acid (HIn, colour A) and conjugate base ($\mathrm{In^-}$, colour B) forms: $\mathrm{HIn(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + In^-(aq)}$.

Common Fluid / Substance	Typical pH	Common Fluid / Substance	Typical pH
Saturated NaOH solution	~ 15	Human saliva	6.4
$0.1\ \mathrm{M\ NaOH}$ solution	13	Black coffee	5.0
Lime water	10.5	Tomato juice	~ 4.2
Milk of magnesia	10	Soft drinks and vinegar	~ 3.0
Egg white, sea water	7.8	Lemon juice	~ 2.2
Human blood	7.4	Gastric juice	~ 1.2
Milk	6.8	$1\ \mathrm{M\ HCl}$ / Conc. HCl	~ 0 / ~ -1.0

7.3 Ionization of Weak Acids (K_a), Weak Bases (K_b), and Polyprotic Acids

For a weak monoprotic acid HX of initial molarity c and degree of ionization α: $\mathrm{HX(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + X^-(aq)}$. At equilibrium, $[\mathrm{HX}] = c(1-\alpha)$ and $[\mathrm{H_3O^+}] = [\mathrm{X^-}] = c\alpha$. Similarly, for a weak monoacidic base $\mathrm{MOH(aq) \rightleftharpoons M^+(aq) + OH^-(aq)}$:

$$K_a = \frac{[\mathrm{H^+}][\mathrm{X^-}]}{[\mathrm{HX}]} = \frac{c\alpha^2}{1-\alpha} \approx c\alpha^2 \quad (\text{if } \alpha \ll 1) \implies \alpha \approx \sqrt{\frac{K_a}{c}}, \quad [\mathrm{H^+}] = c\alpha \approx \sqrt{K_a c}, \quad \mathrm{p}K_a = -\log K_a$$

$$K_b = \frac{[\mathrm{M^+}][\mathrm{OH^-}]}{[\mathrm{MOH}]} = \frac{c\alpha^2}{1-\alpha} \approx c\alpha^2 \quad (\text{if } \alpha \ll 1) \implies \alpha \approx \sqrt{\frac{K_b}{c}}, \quad [\mathrm{OH^-}] = c\alpha \approx \sqrt{K_b c}, \quad \mathrm{p}K_b = -\log K_b$$

This relationship ($\alpha = \sqrt{K_a/c}$) is known as **Ostwald's Dilution Law**: as the concentration c decreases (upon dilution), the degree of ionization α of a weak electrolyte increases. Percent dissociation is defined as $\dfrac{[\mathrm{HA}]_{\text{dissociated}}}{[\mathrm{HA}]_{\text{initial}}} \times 100\% = \alpha \times 100\%$.

Relation Between K_a and K_b for a Conjugate Acid–Base Pair: Adding the ionization of a weak base B and its conjugate acid $\mathrm{BH^+}$ yields the auto-ionization of water ($\mathrm{2H_2O \rightleftharpoons H_3O^+ + OH^-}$):

$$K_a \times K_b = K_w = 1.0 \times 10^{-14} \quad \Longleftrightarrow \quad \mathrm{p}K_a + \mathrm{p}K_b = \mathrm{p}K_w = 14 \quad (\text{at } 298\ \mathrm{K})$$

Di- and Polybasic Acids: Acids with more than one ionizable proton per molecule (such as oxalic acid $\mathrm{H_2C_2O_4}$, $\mathrm{H_2SO_4}$, $\mathrm{H_2CO_3}$, $\mathrm{H_2S}$, citric acid, and $\mathrm{H_3PO_4}$) ionize in successive steps characterized by $K_{a_1}, K_{a_2}, K_{a_3}$. Always:

$$K_{a_1} \gg K_{a_2} \gg K_{a_3}$$

because it is electrostatically much harder to remove a positively charged $\mathrm{H^+}$ ion from a negatively charged ion (such as $\mathrm{HCO_3^-}$ or $\mathrm{H_2PO_4^-}$) or doubly charged ion ($\mathrm{HPO_4^{2-}}$) than from a neutral molecule ($\mathrm{H_2CO_3}$ or $\mathrm{H_3PO_4}$). Consequently, in an aqueous solution of a diprotic acid $\mathrm{H_2A}$, $[\mathrm{H_3O^+}]$ comes almost entirely from the first dissociation step.

Selected Weak Acid ($298\ \mathrm{K}$)	K_a	Selected Weak Base ($298\ \mathrm{K}$)	K_b	Polyprotic Acid ($298\ \mathrm{K}$)	$K_{a_1}, K_{a_2}, (K_{a_3})$
Hydrofluoric (HF)	3.5×10^{-4}	Dimethylamine $(\mathrm{CH_3})_2\mathrm{NH}$	5.4×10^{-4}	Oxalic acid	$5.9 \times 10^{-2},\ 6.4 \times 10^{-5}$
Nitrous ($\mathrm{HNO_2}$)	4.5×10^{-4}	Triethylamine $(\mathrm{C_2H_5})_3\mathrm{N}$	6.45×10^{-5}	Ascorbic acid	$7.4 \times 10^{-4},\ 1.6 \times 10^{-12}$
Formic (HCOOH)	1.8×10^{-4}	Ammonia ($\mathrm{NH_3}$)	1.77×10^{-5}	Sulphurous ($\mathrm{H_2SO_3}$)	$1.7 \times 10^{-2},\ 6.4 \times 10^{-8}$
Benzoic ($\mathrm{C_6H_5COOH}$)	6.5×10^{-5}	Quinine	1.10×10^{-6}	Sulphuric ($\mathrm{H_2SO_4}$)	Very large,\; 1.2×10^{-2}
Acetic ($\mathrm{CH_3COOH}$)	1.74×10^{-5}	Pyridine ($\mathrm{C_5H_5N}$)	1.77×10^{-9}	Carbonic ($\mathrm{H_2CO_3}$)	$4.3 \times 10^{-7},\ 5.6 \times 10^{-11}$
Niacin ($\mathrm{C_5H_4NCOOH}$)	1.5×10^{-5}	Aniline ($\mathrm{C_6H_5NH_2}$)	4.27×10^{-10}	Citric acid	$7.4 \times 10^{-4},\ 1.7 \times 10^{-5},\ 4.0 \times 10^{-7}$
Hypochlorous (HClO)	3.0×10^{-8}	Urea $\mathrm{CO(NH_2)_2}$	1.3×10^{-14}	Phosphoric ($\mathrm{H_3PO_4}$)	$7.5 \times 10^{-3},\ 6.2 \times 10^{-8},\ 4.2 \times 10^{-12}$
Hydrocyanic (HCN) / Phenol	4.9×10^{-10} / 1.3×10^{-10}	–	–	–	–

8. Common Ion Effect, Salt Hydrolysis, Buffers, and Solubility Product (K_{sp})

8.1 Common Ion Effect in Weak Electrolytes

The **common ion effect** is a shift in an ionic equilibrium caused by adding a strong electrolyte that supplies an ionic species already present in the dissociation equilibrium. For example, adding $\mathrm{CH_3COONa}$ (or HCl) to $\mathrm{CH_3COOH(aq) \rightleftharpoons H^+(aq) + CH_3COO^-(aq)}$ suppresses the degree of ionization of acetic acid in accordance with Le Chatelier's principle.

8.2 Hydrolysis of Salts and pH of Salt Solutions

Hydrolysis is the interaction of cations, anions, or both of a salt with water to reform the parent weak acid or weak base. We classify salts into four types (at $298\ \mathrm{K}$):

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Salt Category & Examples	Hydrolyzing Ion & Reaction	Nature of Solution & pH Formula (298 K)
1. Strong Acid + Strong Base ($NaCl, KBr, NaNO_3, KClO_4$)	Neither cation ($Na^+, K^+, Ca^{2+}, Ba^{2+}$) nor anion ($Cl^-, Br^-, NO_3^-, ClO_4^-$) hydrolyzes; ions only get hydrated.	**Neutral:** $pH = 7.0$
2. Weak Acid + Strong Base ($CH_3COONa, NaCN, NaNO_2, KF$)	Anion hydrolyzes: $A^-(aq) + H_2O(l) \rightleftharpoons HA(aq) + OH^-(aq)$ $K_h = \frac{K_w}{K_a}, \quad h = \sqrt{\frac{K_w}{K_a c}}$	**Basic (Alkaline):** $pH > 7$ $pH = 7 + \frac{1}{2}(pK_a + \log c)$
3. Strong Acid + Weak Base ($NH_4Cl, NH_4NO_3, C_5H_5NH^+Cl^-$)	Cation hydrolyzes: $BH^+(aq) + H_2O(l) \rightleftharpoons B(aq) + H_3O^+(aq)$ $K_h = \frac{K_w}{K_b}, \quad h = \sqrt{\frac{K_w}{K_b c}}$	**Acidic:** $pH < 7$ $pH = 7 - \frac{1}{2}(pK_b + \log c)$
4. Weak Acid + Weak Base (CH_3COONH_4)	Both ions hydrolyze: $A^- + BH^+ + H_2O \rightleftharpoons HA + BOH$. Degree of hydrolysis is independent of concentration c.	$pH = 7 + \frac{1}{2}(pK_a - pK_b)$ ($pH > 7$ if $pK_a > pK_b$; $pH < 7$ if $pK_a < pK_b$)

8.3 Buffer Solutions and the Henderson–Hasselbalch Equation

Solutions that resist changes in pH upon dilution or upon addition of small amounts of acid or alkali are called **buffer solutions**. Human blood is naturally buffered around pH 7.4.

- **Acidic Buffer** (weak acid HA + its salt with a strong base, e.g., $CH_3COOH + CH_3COONa$, buffer around pH 4.76):

$$pH = pK_a + \log \frac{[\text{Conjugate base, } A^-]}{[\text{Acid, } HA]} \approx pK_a + \log \frac{[\text{Salt}]}{[\text{Acid}]}$$

- **Basic Buffer** (weak base B + its salt with a strong acid, e.g., $NH_3 + NH_4Cl$, buffer around pH 9.25):

$$pOH = pK_b + \log \frac{[\text{Conjugate acid, } BH^+]}{[\text{Base, } B]} \quad \Longleftrightarrow \quad pH = pK_a(\text{of } BH^+) - \log \frac{[BH^+]}{[B]}$$

(Note: Using $pK_w - pH = pK_w - pK_a + \log \frac{[BH^+]}{[B]}$, when $[BH^+] = [B]$, $pOH = pK_b = 4.75$ and $pH = pK_a = 9.25$. Dilution does not change buffer pH because the concentration ratio inside the logarithm remains constant.)

8.4 Solubility Equilibria of Sparingly Soluble Salts (K_{sp})

A salt dissolves in a solvent only if its solvation enthalpy (always negative) overcomes its lattice enthalpy. Salts are classified by molar solubility S at 298 K into: **Category I (Soluble):** $S > 0.1$ M; **Category II (Slightly Soluble):** 0.01 M $< S < 0.1$ M; and **Category III (Sparingly Soluble):** $S < 0.01$ M.

For a general sparingly soluble salt $M_xX_y(s) \rightleftharpoons xM^{p+}(aq) + yX^{q-}(aq)$ (where $xp^+ = yq^-$) with molar solubility S (mol L^{-1}):

$$K_{sp} = [M^{p+}]^x[X^{q-}]^y = (xS)^x(yS)^y = x^x \cdot y^y \cdot S^{(x+y)} \implies S = \left(\frac{K_{sp}}{x^x \cdot y^y}\right)^{\frac{1}{x+y}}$$

- $1:1$ salt (AB, e.g., $AgCl, BaSO_4, AgCN$): $K_{sp} = S^2 \implies S = \sqrt{K_{sp}}$
- $1:2$ or $2:1$ salt (AB_2 or A_2B, e.g., $PbCl_2, Ni(OH)_2, Ag_2CrO_4, Hg_2I_2$ [which dissociates as $Hg_2^{2+} + 2I^-$]): $K_{sp} = 4S^3 \implies S = (K_{sp}/4)^{1/3}$
- $1:3$ salt (AB_3, e.g., $Fe(OH)_3, Al(OH)_3$): $K_{sp} = 27S^4 \implies S = (K_{sp}/27)^{1/4}$
- $2:3$ salt (A_2X_3): $K_{sp} = (2S)^2(3S)^3 = 108S^5$; $\quad 3:4$ salt ($Zr_3(PO_4)_4$): $K_{sp} = (3S)^3(4S)^4 = 6912S^7$

Salt Name & Formula	K_{sp} (298 K)	Salt Name & Formula	K_{sp} (298 K)	Salt Name & Formula	K_{sp} (298 K)
Silver Chloride ($AgCl$)	1.8×10^{-10}	Barium Sulphate ($BaSO_4$)	1.1×10^{-10}	Ferric Hydroxide ($Fe(OH)_3$)	1.0×10^{-38}
Silver Bromide ($AgBr$)	5.0×10^{-13}	Barium Chromate ($BaCrO_4$)	1.2×10^{-10}	Ferrous Sulphide (FeS)	6.3×10^{-18}
Silver Iodide (AgI)	8.3×10^{-17}	Calcium Sulphate ($CaSO_4$)	9.1×10^{-6}	Manganese Sulphide (MnS)	2.5×10^{-13}
Silver Chromate (Ag_2CrO_4)	1.1×10^{-12}	Calcium Fluoride (CaF_2)	5.3×10^{-9}	Zinc Sulphide (ZnS)	1.6×10^{-24}
Silver Carbonate (Ag_2CO_3)	8.1×10^{-12}	Lead Chloride ($PbCl_2$)	1.6×10^{-5}	Cadmium Sulphide (CdS)	8.0×10^{-27}
Aluminium Hydroxide ($Al(OH)_3$)	1.3×10^{-33}	Mercurous Iodide (Hg_2I_2)	4.5×10^{-29}	Nickel Hydroxide ($Ni(OH)_2$)	2.0×10^{-15}

Criteria for Precipitation & Common Ion Effect on Solubility:

- If ionic product $Q_{sp} > K_{sp}$, precipitation occurs until $Q_{sp} = K_{sp}$. If $Q_{sp} < K_{sp}$, the solution is unsaturated (no precipitation).
- Adding a common ion decreases the molar solubility of a sparingly soluble salt (e.g., passing HCl gas through saturated $NaCl$ solution precipitates high-purity $NaCl$; gravimetric estimation of Ag^+ as $AgCl$, Fe^{3+} as $Fe(OH)_3$, and Ba^{2+} as $BaSO_4$).
- **Effect of pH on Solubility of Salts of Weak Acids (MX):** For a salt $MX(s) \rightleftharpoons M^+(aq) + X^-(aq)$ where HX is a weak acid with dissociation constant K_a, lowering the pH (increasing $[H^+]$) protonates X^- to HX. The fraction of anion remaining unprotonated as X^- is

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$f = \dfrac{[\mathrm{X}^-]}{[\mathrm{X}^-] + [\mathrm{HX}]} = \dfrac{K_a}{K_a + [\mathrm{H}^+]}$. Since $K_{\mathrm{sp}} = [\mathrm{M}^+][\mathrm{X}^-] = (S)(fS) = S^2\left(\dfrac{K_a}{K_a + [\mathrm{H}^+]}\right)$, the molar solubility S at any pH is given by:

$$S = \left[\frac{K_{\mathrm{sp}}([\mathrm{H}^+] + K_a)}{K_a}\right]^{1/2} = \sqrt{K_{\mathrm{sp}}\left(1 + \frac{[\mathrm{H}^+]}{K_a}\right)}$$

Thus, solubility S increases as $[\mathrm{H}^+]$ increases (pH decreases).

9. Step-by-Step Worked Examples

Worked Example 1: Calculating K_p, K_c, and Equilibrium Partial Pressures from Total Pressure

Problem: $13.8\,\mathrm{g}$ of $\mathrm{N_2O_4}$ was placed in a $1\,\mathrm{L}$ reaction vessel at $400\,\mathrm{K}$ and allowed to attain the equilibrium $\mathrm{N_2O_4(g)} \rightleftharpoons 2\mathrm{NO_2(g)}$. The total pressure at equilibrium was found to be $9.15\,\mathrm{bar}$. Calculate K_p, K_c, and the partial pressures at equilibrium. (Given: Molar mass of $\mathrm{N_2O_4} = 92\,\mathrm{g\,mol^{-1}}$, $R = 0.083\,\mathrm{bar\,L\,mol^{-1}\,K^{-1}}$).

Solution:

Step 1: Find initial pressure of $\mathrm{N_2O_4}$ using the ideal gas law.

Number of moles of $\mathrm{N_2O_4}$, $n = \dfrac{13.8\,\mathrm{g}}{92\,\mathrm{g\,mol^{-1}}} = 0.15\,\mathrm{mol}$.

Initial pressure $p_0 = \dfrac{nRT}{V} = \dfrac{0.15\,\mathrm{mol} \times 0.083\,\mathrm{bar\,L\,mol^{-1}\,K^{-1}} \times 400\,\mathrm{K}}{1\,\mathrm{L}} = 4.98\,\mathrm{bar}$.

Step 2: Set up the ICE table in terms of partial pressures (bar).

For $\mathrm{N_2O_4(g)} \rightleftharpoons 2\mathrm{NO_2(g)}$, let x bar be the decrease in partial pressure of $\mathrm{N_2O_4}$. At equilibrium:

$p_{\mathrm{N_2O_4}} = (4.98 - x)$ bar and $p_{\mathrm{NO_2}} = 2x$ bar.

Total pressure at equilibrium: $p_{\mathrm{total}} = (4.98 - x) + 2x = 4.98 + x = 9.15\,\mathrm{bar} \implies x = 9.15 - 4.98 = 4.17\,\mathrm{bar}$.

Therefore, $p_{\mathrm{N_2O_4}} = 4.98 - 4.17 = \mathbf{0.81}$ bar and $p_{\mathrm{NO_2}} = 2 \times 4.17 = \mathbf{8.34}$ bar.

Step 3: Calculate K_p and K_c.

$$K_p = \frac{(p_{\mathrm{NO_2}})^2}{p_{\mathrm{N_2O_4}}} = \frac{(8.34)^2}{0.81} = \mathbf{85.87}$$

Since $\Delta n = 2 - 1 = 1$ and $K_p = K_c(RT)^{\Delta n}$:

$$85.87 = K_c(0.083 \times 400)^1 \implies K_c = \frac{85.87}{33.2} = 2.586 \approx \mathbf{2.6}$$

Worked Example 2: pH of an Extremely Dilute Strong Acid Solution

Problem: Calculate the pH of a $1.0 \times 10^{-8}\,\mathrm{M}$ aqueous solution of HCl at $298\,\mathrm{K}$.

Solution:

If we naively wrote $\mathrm{pH} = -\log(10^{-8}) = 8$, an acidic solution would appear basic, which is physically impossible! In very dilute acid solutions ($[\mathrm{HA}] < 10^{-6}\,\mathrm{M}$), the $\mathrm{H_3O^+}$ contribution from the auto-ionization of water cannot be neglected.

Let $x = [\mathrm{OH^-}]$ generated from $2\mathrm{H_2O(l)} \rightleftharpoons \mathrm{H_3O^+(aq)} + \mathrm{OH^-(aq)}$. Since HCl is completely ionized ($[\mathrm{H_3O^+}]_{\mathrm{HCl}} = 10^{-8}\,\mathrm{M}$), the total hydronium concentration at equilibrium is:

$$[\mathrm{H_3O^+}] = (10^{-8} + x)\,\mathrm{M} \quad \text{and} \quad [\mathrm{OH^-}] = x\,\mathrm{M}$$

Substituting into $K_w = [\mathrm{H_3O^+}][\mathrm{OH^-}] = 10^{-14}$:

$$(10^{-8} + x)(x) = 10^{-14} \implies x^2 + 10^{-8}x - 10^{-14} = 0$$

Solving the quadratic equation gives $x = \dfrac{-10^{-8} + \sqrt{10^{-16} + 4 \times 10^{-14}}}{2} = 9.5 \times 10^{-8}\,\mathrm{M} = [\mathrm{OH^-}]$.

Therefore, $[\mathrm{H_3O^+}] = 10^{-8} + 9.5 \times 10^{-8} = 1.05 \times 10^{-7}\,\mathrm{M}$, or $\mathrm{pOH} = -\log(9.5 \times 10^{-8}) = 7.02$, giving $\mathrm{pH} = 14 - 7.02 = \mathbf{6.98}$.

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Worked Example 3: Partial Neutralization of a Weak Base to Form a Basic Buffer

Problem: Calculate (a) the pH of a $0.10\,\text{M}$ ammonia (NH_3) solution, and (b) the pH after $50.0\,\text{mL}$ of this solution is treated with $25.0\,\text{mL}$ of $0.10\,\text{M}$ HCl. Given: $K_b(NH_3) = 1.77 \times 10^{-5}$ at $298\,\text{K}$.

Solution:

Part (a) Before neutralization: For $NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$, let $[NH_4^+] = [OH^-] = x$. Since $x \ll 0.10$:

$$K_b = \frac{x^2}{0.10} = 1.77 \times 10^{-5} \implies x = [OH^-] = 1.33 \times 10^{-3}\,\text{M}$$

$$[H^+] = \frac{10^{-14}}{1.33 \times 10^{-3}} = 7.51 \times 10^{-12}\,\text{M} \implies \text{pH} = -\log(7.51 \times 10^{-12}) = \mathbf{11.12}$$

Part (b) After adding $25.0\,\text{mL}$ of $0.10\,\text{M}$ HCl:

Initial millimoles of $NH_3 = 50.0\,\text{mL} \times 0.10\,\text{M} = 5.0\,\text{mmol}$.

Added millimoles of $HCl = 25.0\,\text{mL} \times 0.10\,\text{M} = 2.5\,\text{mmol}$.

The neutralization reaction $NH_3 + HCl \longrightarrow NH_4^+ + Cl^-$ consumes $2.5\,\text{mmol}$ of NH_3 and produces $2.5\,\text{mmol}$ of NH_4^+ in a total volume of $50.0 + 25.0 = 75.0\,\text{mL}$:

$$[NH_3]_{\text{remaining}} = \frac{2.5\,\text{mmol}}{75.0\,\text{mL}} = 0.033\,\text{M}, \qquad [NH_4^+]_{\text{formed}} = \frac{2.5\,\text{mmol}}{75.0\,\text{mL}} = 0.033\,\text{M}$$

Substituting into the base ionization equilibrium $K_b = \dfrac{[NH_4^+][OH^-]}{[NH_3]}$:

$$1.77 \times 10^{-5} = \frac{(0.033)[OH^-]}{(0.033)} \implies [OH^-] = 1.77 \times 10^{-5}\,\text{M}$$

$$[H^+] = \frac{10^{-14}}{1.77 \times 10^{-5}} = 5.65 \times 10^{-10}\,\text{M} \implies \text{pH} = -\log(5.65 \times 10^{-10}) = \mathbf{9.25}\ (\text{or } 9.24)$$

Worked Example 4: Solubility of a Salt of a Weak Acid in a Buffer Solution

Problem: The ionization constant of benzoic acid (C_6H_5COOH) is $K_a = 6.46 \times 10^{-5}$ and K_{sp} for silver benzoate (C_6H_5COOAg) is 2.5×10^{-13}. How many times is silver benzoate more soluble in a buffer of pH 3.19 compared to its solubility in pure water?

Solution:

Step 1: Solubility in pure water (S_0).

Silver benzoate is a $1:1$ salt ($C_6H_5COOAg(s) \rightleftharpoons C_6H_5COO^-(aq) + Ag^+(aq)$):

$$S_0 = \sqrt{K_{sp}} = \sqrt{2.5 \times 10^{-13}} = 5.0 \times 10^{-7}\,\text{mol L}^{-1}$$

Step 2: Solubility in a buffer of pH 3.19 (S_{buffer}).

At pH $= 3.19$, $[H^+] = 10^{-3.19} = 6.46 \times 10^{-4}\,\text{M}$.

Using the pH-dependent solubility equation $S_{\text{buffer}} = \sqrt{K_{sp}\left(\dfrac{[H^+] + K_a}{K_a}\right)}$:

$$\frac{S_{\text{buffer}}}{S_0} = \sqrt{\frac{[H^+] + K_a}{K_a}} = \sqrt{\frac{6.46 \times 10^{-4} + 6.46 \times 10^{-5}}{6.46 \times 10^{-5}}} = \sqrt{10 + 1} = \sqrt{11} \approx \mathbf{3.32}$$

Thus, silver benzoate is 3.32 times more soluble in the pH 3.19 buffer than in pure water.

10. Exam-Aligned Practice Questions (NEET UG Standard)

Note: All questions below are original practice items authored strictly in accordance with the NEET UG examination format and syllabus.

BOSONANCE

Practice Question 1 (Single-Correct MCQ • Equilibrium Constant Manipulation)

At a certain temperature T, the equilibrium constant K_c for the reaction $\mathrm{N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)}$ is $64\,\mathrm{L^2\,mol^{-2}}$. What will be the value of the equilibrium constant K_c' for the decomposition reaction $\mathrm{NH_3(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{3}{2}H_2(g)}$ at the same temperature?

(1) $8.0\,\mathrm{mol\,L^{-1}}$

(2) $0.125\,\mathrm{mol\,L^{-1}}$

(3) $0.0156\,\mathrm{mol\,L^{-1}}$

(4) $32.0\,\mathrm{mol\,L^{-1}}$

Correct Answer: (2) $0.125\,\mathrm{mol\,L^{-1}}$

Worked Solution: The target equation $\mathrm{NH_3(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{3}{2}H_2(g)}$ is obtained by reversing the given synthesis equation and multiplying its stoichiometric coefficients by $\frac{1}{2}$.

Reversing the reaction gives $K_{\text{rev}} = \dfrac{1}{K_c}$. Multiplying by $\frac{1}{2}$ raises the constant to the power $\frac{1}{2}$:

$$K_c' = \left(\frac{1}{K_c}\right)^{1/2} = \frac{1}{\sqrt{64}} = \frac{1}{8} = 0.125\,\mathrm{mol\,L^{-1}}$$

Practice Question 2 (Single-Correct MCQ • K_p and K_c Relationship)

For the heterogeneous reaction $\mathrm{FeO(s) + CO(g) \rightleftharpoons Fe(s) + CO_2(g)}$ at $1050\,\mathrm{K}$, $K_p = 0.265$. If the initial partial pressures in a closed vessel at $1050\,\mathrm{K}$ are $p_{\mathrm{CO}} = 1.40\,\mathrm{atm}$ and $p_{\mathrm{CO_2}} = 0.80\,\mathrm{atm}$, which statement correctly describes the system?

(1) $Q_p = 0.571$; net reaction proceeds in the reverse direction until $p_{\mathrm{CO}} = 1.74\,\mathrm{atm}$ and $p_{\mathrm{CO_2}} = 0.46\,\mathrm{atm}$.

(2) $Q_p = 0.571$; net reaction proceeds in the forward direction until $p_{\mathrm{CO}} = 1.06\,\mathrm{atm}$ and $p_{\mathrm{CO_2}} = 1.14\,\mathrm{atm}$.

(3) $Q_p = 1.75$; net reaction proceeds in the reverse direction until $p_{\mathrm{CO}} = 1.74\,\mathrm{atm}$ and $p_{\mathrm{CO_2}} = 0.46\,\mathrm{atm}$.

(4) Since $\mathrm{FeO(s)}$ and $\mathrm{Fe(s)}$ are solids, $K_p \neq K_c$ and the reaction is already at equilibrium.

Correct Answer: (1) $Q_p = 0.571$**; net reaction proceeds in the reverse direction until** $p_{\mathrm{CO}} = 1.74\,\mathrm{atm}$ **and** $p_{\mathrm{CO_2}} = 0.46\,\mathrm{atm}$.

Worked Solution: Pure solids $\mathrm{FeO(s)}$ and $\mathrm{Fe(s)}$ are omitted from Q_p and K_p:

$$Q_p = \frac{p_{\mathrm{CO_2}}}{p_{\mathrm{CO}}} = \frac{0.80}{1.40} = 0.571$$

Since $Q_p(0.571) > K_p(0.265)$, the net reaction proceeds in the **reverse direction** (forming CO and consuming $\mathrm{CO_2}$). Let $x\,\mathrm{atm}$ be the decrease in $p_{\mathrm{CO_2}}$:

$$\frac{0.80 - x}{1.40 + x} = 0.265 \implies 0.80 - x = 0.371 + 0.265x \implies 1.265x = 0.429 \implies x = 0.339 \approx 0.34\,\mathrm{atm}$$

Therefore, at equilibrium: $p_{\mathrm{CO}} = 1.40 + 0.34 = 1.74\,\mathrm{atm}$ and $p_{\mathrm{CO_2}} = 0.80 - 0.34 = 0.46\,\mathrm{atm}$.

BOSONANCE

Practice Question 3 (Single-Correct MCQ • Degree of Dissociation and K_p)

At a certain temperature and a total equilibrium pressure of $1.0\ \text{bar}$ ($10^5\ \text{Pa}$), iodine vapour contains 40% by volume of iodine atoms (I) according to the equilibrium $\text{I}_2(\text{g}) \rightleftharpoons 2\text{I}(\text{g})$. What is the value of K_p (in bar) for this equilibrium?

(1) $0.67\ \text{bar}$

(2) $0.267\ \text{bar}$

(3) $1.50\ \text{bar}$

(4) $0.16\ \text{bar}$

Correct Answer: (2) $0.267\ \text{bar}$

Worked Solution: For gases at the same temperature and pressure, volume percentage equals mole percentage (mole fraction $\times 100$). Thus, mole fraction of $\text{I}(\text{g})$ is $x_{\text{I}} = 0.40$ and mole fraction of $\text{I}_2(\text{g})$ is $x_{\text{I}_2} = 1 - 0.40 = 0.60$.
At total pressure $P_{\text{total}} = 1.0\ \text{bar}$:

$$p_{\text{I}} = x_{\text{I}} P_{\text{total}} = 0.40\ \text{bar}, \qquad p_{\text{I}_2} = x_{\text{I}_2} P_{\text{total}} = 0.60\ \text{bar}$$

$$K_p = \frac{(p_{\text{I}})^2}{p_{\text{I}_2}} = \frac{(0.40)^2}{0.60} = \frac{0.16}{0.60} = 0.267\ \text{bar} \quad (\text{or } 2.67 \times 10^4\ \text{Pa})$$

Practice Question 4 (Single-Correct MCQ • Thermodynamics of Equilibrium)

For the reaction $\text{NO}(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightleftharpoons \text{NO}_2(\text{g})$ at $298\ \text{K}$, the standard molar Gibbs energies of formation are $\Delta_f G^\circ(\text{NO}_2) = 52.0\ \text{kJ mol}^{-1}$, $\Delta_f G^\circ(\text{NO}) = 87.0\ \text{kJ mol}^{-1}$, and $\Delta_f G^\circ(\text{O}_2) = 0\ \text{kJ mol}^{-1}$. What is $\Delta_r G^\circ$ and the approximate order of magnitude of the equilibrium constant K at $298\ \text{K}$? (Take $2.303\,RT = 5.705\ \text{kJ mol}^{-1}$ at $298\ \text{K}$).

(1) $\Delta_r G^\circ = -35.0\ \text{kJ mol}^{-1}, \quad K \approx 1.38 \times 10^6$

(2) $\Delta_r G^\circ = +35.0\ \text{kJ mol}^{-1}, \quad K \approx 7.2 \times 10^{-7}$

(3) $\Delta_r G^\circ = -139.0\ \text{kJ mol}^{-1}, \quad K \approx 2.3 \times 10^{24}$

(4) $\Delta_r G^\circ = -35.0\ \text{kJ mol}^{-1}, \quad K \approx 6.13$

Correct Answer: (1) $\Delta_r G^\circ = -35.0\ \text{kJ mol}^{-1}, \quad K \approx 1.38 \times 10^6$

Worked Solution:

$$\Delta_r G^\circ = \Delta_f G^\circ(\text{NO}_2) - \left[\Delta_f G^\circ(\text{NO}) + \frac{1}{2}\Delta_f G^\circ(\text{O}_2)\right] = 52.0 - (87.0 + 0) = -35.0\ \text{kJ mol}^{-1}$$

Using $\Delta_r G^\circ = -2.303\,RT \log K$:

$$\log K = \frac{-\Delta_r G^\circ}{2.303\,RT} = \frac{35.0\ \text{kJ mol}^{-1}}{5.705\ \text{kJ mol}^{-1}} = 6.135 \implies K = 10^{6.135} \approx 1.38 \times 10^6$$

BOSONANCE

Practice Question 5 (Assertion–Reason • Le Chatelier's Principle)

Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A): For the endothermic steam-reforming reaction $\mathrm{CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)}$, increasing the temperature increases the value of K_p, whereas adding a catalyst increases the equilibrium yield of $\mathrm{H_2(g)}$ without changing K_p.
Reason (R): A catalyst lowers the activation energy for both the forward and reverse reactions by the exact same amount and does not alter the equilibrium composition of a reaction mixture.
In the light of the above statements, choose the correct option:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

Correct Answer: (4) (A) is false but (R) is true.

Worked Solution: Because the reaction is endothermic ($\Delta H > 0$), increasing temperature does increase K_p. However, adding a catalyst **does not** increase the equilibrium yield of $\mathrm{H_2(g)}$; it only helps the system attain the exact same equilibrium composition faster. Therefore, Assertion (A) is false and Reason (R) is true.

Practice Question 6 (Matching Type • Brönsted–Lowry & Lewis Acid–Base Concepts)

Match each chemical species in **Column I** with its correct acid–base description in **Column II**:

Column I (Species)	Column II (Acid–Base Role / Property)
(A) $\mathrm{HCO_3^-}$	(i) Electron-deficient Lewis acid lacking any ionizable proton
(B) $\mathrm{BCl_3}$	(ii) Brönsted acid only; cannot act as a Lewis acid due to absence of vacant valence orbitals
(C) $\mathrm{NH_4^+}$	(iii) Amphoteric species with conjugate acid $\mathrm{H_2CO_3}$ and conjugate base $\mathrm{CO_3^{2-}}$
(D) $\mathrm{NH_2^-}$	(iv) Very strong Brönsted base whose conjugate acid is $\mathrm{NH_3}$

(1) (A)-(iii), (B)-(i), (C)-(ii), (D)-(iv)

(2) (A)-(iii), (B)-(ii), (C)-(i), (D)-(iv)

(3) (A)-(iv), (B)-(i), (C)-(ii), (D)-(iii)

(4) (A)-(i), (B)-(iii), (C)-(iv), (D)-(ii)

Correct Answer: (1) (A)-(iii), (B)-(i), (C)-(ii), (D)-(iv)

Worked Solution:

- (A) $\mathrm{HCO_3^-}$ is amphoteric: gaining $\mathrm{H^+}$ gives its conjugate acid $\mathrm{H_2CO_3}$, and losing $\mathrm{H^+}$ gives its conjugate base $\mathrm{CO_3^{2-}} \rightarrow$ (iii).
- (B) $\mathrm{BCl_3}$ has an incomplete octet (6 valence electrons around B) and accepts a lone pair, acting as an aprotic Lewis acid $\rightarrow$ (i).
- (C) $\mathrm{NH_4^+}$ can donate $\mathrm{H^+}$ (Brönsted acid), but nitrogen has a complete octet and no vacant orbital to accept an electron pair $\rightarrow$ (ii).
- (D) $\mathrm{NH_2^-}$ gains one proton to form its conjugate acid $\mathrm{NH_3} \rightarrow$ (iv).

BOSONANCE

Practice Question 7 (Single-Correct MCQ • Weak Acid Ionization & Common Ion Effect)

The ionization constant of phenol (C_6H_5OH) at $298\,K$ is 1.0×10^{-10}. What is (i) the concentration of phenolate ion ($C_6H_5O^-$) in a $0.05\,M$ aqueous solution of phenol, and (ii) the degree of ionization (α) of phenol if the solution is also made $0.01\,M$ in sodium phenolate (C_6H_5ONa)?

(1) (i) $2.24 \times 10^{-6}\,M$, (ii) 1.0×10^{-8}

(2) (i) $5.0 \times 10^{-12}\,M$, (ii) 5.0×10^{-9}

(3) (i) $2.24 \times 10^{-6}\,M$, (ii) 4.48×10^{-5}

(4) (i) $1.0 \times 10^{-5}\,M$, (ii) 1.0×10^{-8}

Correct Answer: (1) (i) $2.24 \times 10^{-6}\,M$, (ii) 1.0×10^{-8}

Worked Solution:

Part (i): In $0.05\,M$ pure phenol, $[C_6H_5O^-] = [H^+] = x$:

$$K_a = \frac{x^2}{0.05 - x} \approx \frac{x^2}{0.05} = 1.0 \times 10^{-10} \implies x^2 = 5.0 \times 10^{-12} \implies x = 2.24 \times 10^{-6}\,M$$

Part (ii): In the presence of $0.01\,M$ C_6H_5ONa, $[C_6H_5O^-] \approx 0.01\,M$ and $[C_6H_5OH] = 0.05(1-\alpha) \approx 0.05\,M$, while $[H^+] = 0.05\alpha$:

$$K_a = \frac{[H^+][C_6H_5O^-]}{[C_6H_5OH]} = \frac{(0.05\alpha)(0.01)}{0.05} = 0.01\alpha = 1.0 \times 10^{-10} \implies \alpha = 1.0 \times 10^{-8}$$

Practice Question 8 (Single-Correct MCQ • Diprotic Acid H_2S Equilibria)

For H_2S in aqueous solution at $298\,K$, the first ionization constant is $K_{a_1} = 9.1 \times 10^{-8}$ and the second ionization constant is $K_{a_2} = 1.2 \times 10^{-13}$. Compare the concentration of sulphide ion $[S^{2-}]$ in (i) a pure $0.10\,M$ H_2S solution, and (ii) a $0.10\,M$ H_2S solution that also contains $0.10\,M$ HCl.

(1) (i) $9.54 \times 10^{-5}\,M$, (ii) $9.1 \times 10^{-8}\,M$

(2) (i) $1.2 \times 10^{-13}\,M$, (ii) $1.09 \times 10^{-19}\,M$

(3) (i) $1.2 \times 10^{-13}\,M$, (ii) $1.2 \times 10^{-13}\,M$

(4) (i) $1.09 \times 10^{-19}\,M$, (ii) $1.2 \times 10^{-13}\,M$

Correct Answer: (2) (i) $1.2 \times 10^{-13}\,M$, (ii) $1.09 \times 10^{-19}\,M$

Worked Solution:

Case (i) Pure $0.10\,M$ H_2S solution: Since $K_{a_1} \gg K_{a_2}$, almost all H^+ and HS^- come from the first step ($H_2S \rightleftharpoons H^+ + HS^-$), so $[H^+] \approx [HS^-] = \sqrt{K_{a_1} c} = \sqrt{9.1 \times 10^{-8} \times 0.10} = 9.54 \times 10^{-5}\,M$.

Substituting $[H^+] = [HS^-]$ into the second ionization constant $K_{a_2} = \dfrac{[H^+][S^{2-}]}{[HS^-]}$ immediately gives:

$$[S^{2-}] = K_{a_2} = 1.2 \times 10^{-13}\,M$$

Case (ii) In the presence of $0.10\,M$ HCl: Now $[H^+] = 0.10\,M$ (from strong acid HCl) and $[H_2S] \approx 0.10\,M$. Using the overall dissociation constant $K_a = K_{a_1} \times K_{a_2} = \dfrac{[H^+]^2[S^{2-}]}{[H_2S]}$:

$$(9.1 \times 10^{-8})(1.2 \times 10^{-13}) = \frac{(0.10)^2[S^{2-}]}{0.10} \implies [S^{2-}] = \frac{1.092 \times 10^{-20}}{0.10} = 1.09 \times 10^{-19}\,M$$

BOSONANCE

Practice Question 9 (Single-Correct MCQ • Salt Hydrolysis)

The ionization constant of nitrous acid (HNO_2) at $298\,K$ is $K_a = 4.5 \times 10^{-4}$. What is the degree of hydrolysis (h) and the pH of a $0.04\,M$ sodium nitrite ($NaNO_2$) solution at $298\,K$? (Given: $\log 2 = 0.30, \log 3 = 0.48$).

(1) $h = 2.36 \times 10^{-5}, \quad pH = 7.98$

(2) $h = 2.36 \times 10^{-5}, \quad pH = 6.02$

(3) $h = 5.55 \times 10^{-10}, \quad pH = 8.42$

(4) $h = 1.50 \times 10^{-4}, \quad pH = 7.98$

Correct Answer: (1) $h = 2.36 \times 10^{-5}, \quad pH = 7.98$

Worked Solution: $NaNO_2$ is a salt of a weak acid (HNO_2) and a strong base ($NaOH$), so its aqueous solution is basic ($pH > 7$). Anion hydrolysis is $NO_2^-(aq) + H_2O(l) \rightleftharpoons HNO_2(aq) + OH^-(aq)$:

$$K_h = \frac{K_w}{K_a} = \frac{1.0 \times 10^{-14}}{4.5 \times 10^{-4}} = 2.22 \times 10^{-11}$$

$$h = \sqrt{\frac{K_h}{c}} = \sqrt{\frac{2.22 \times 10^{-11}}{0.04}} = \sqrt{5.55 \times 10^{-10}} = 2.36 \times 10^{-5}$$

$$[OH^-] = ch = 0.04 \times 2.36 \times 10^{-5} = 9.44 \times 10^{-7}\,M \implies pOH = -\log(9.44 \times 10^{-7}) = 6.02$$

Therefore, $pH = 14 - 6.02 = 7.98$.

Practice Question 10 (Multi-Statement Evaluation • Acid–Base Mixtures & Neutral Water)

Evaluate the following three statements and choose the correct option:
Statement I: The ionic product of water at $310\,K$ is $2.7 \times 10^{-14}\,M^2$; therefore, neutral water at $310\,K$ has $pH \approx 6.78$.
Statement II: Mixing $10\,mL$ of $0.01\,M$ H_2SO_4 with $10\,mL$ of $0.01\,M$ $Ca(OH)_2$ at $298\,K$ results in a neutral solution with $pH = 7.0$.
Statement III: Mixing $10\,mL$ of $0.1\,M$ H_2SO_4 with $10\,mL$ of $0.1\,M$ KOH at $298\,K$ results in a neutral solution with $pH = 7.0$.

(1) Only Statements I and II are correct.

(2) Only Statements II and III are correct.

(3) All three Statements I, II, and III are correct.

(4) Only Statement I is correct.

Correct Answer: (1) Only Statements I and II are correct.

Worked Solution:

- **Statement I is correct:** In neutral water at $310\,K$, $[H^+] = [OH^-] = \sqrt{K_w} = \sqrt{2.7 \times 10^{-14}} = 1.643 \times 10^{-7}\,M$, so $pH = -\log(1.643 \times 10^{-7}) = 6.78$.
- **Statement II is correct:** H_2SO_4 supplies $2 \times 10 \times 0.01 = 0.2\,mmol$ of H^+, and $Ca(OH)_2$ supplies $2 \times 10 \times 0.01 = 0.2\,mmol$ of OH^-. Exact neutralization occurs ($pH = 7.0$).
- **Statement III is incorrect:** $10\,mL$ of $0.1\,M$ H_2SO_4 supplies $2 \times 10 \times 0.1 = 2.0\,mmol$ of H^+, whereas $10\,mL$ of $0.1\,M$ KOH supplies only $1.0\,mmol$ of OH^-. Thus $1.0\,mmol$ of H^+ remains in $20\,mL$ ($[H^+] = 0.05\,M$, $pH = 1.30$, strongly acidic).

BOSONANCE

Practice Question 11 (Single-Correct MCQ • Comparing Molar Solubilities Across Salt Stoichiometries)

At $298\ \mathrm{K}$, the solubility product constants of $\mathrm{Ag_2CrO_4}$ and AgBr are 1.1×10^{-12} and 5.0×10^{-13}, respectively. What is the approximate ratio of the molarity of a saturated solution of $\mathrm{Ag_2CrO_4}$ (S_1) to that of a saturated solution of AgBr (S_2)?

(1) $2.2 : 1$

(2) $91.9 : 1$

(3) $0.011 : 1$

(4) $1.48 : 1$

Correct Answer: (2) $91.9 : 1$

Worked Solution:

- $\mathrm{Ag_2CrO_4(s) \rightleftharpoons 2Ag^+(aq) + CrO_4^{2-}(aq)}$ is a $2:1$ salt:

$$K_{sp}(\mathrm{Ag_2CrO_4}) = (2S_1)^2(S_1) = 4S_1^3 = 1.1 \times 10^{-12} \implies S_1 = \left(\frac{1.1 \times 10^{-12}}{4}\right)^{1/3} = (2.75 \times 10^{-13})^{1/3} = 6.50 \times 10^{-5}\ \mathrm{M}$$

- $\mathrm{AgBr(s) \rightleftharpoons Ag^+(aq) + Br^-(aq)}$ is a $1:1$ salt:

$$K_{sp}(\mathrm{AgBr}) = S_2^2 = 5.0 \times 10^{-13} \implies S_2 = \sqrt{5.0 \times 10^{-13}} = 7.07 \times 10^{-7}\ \mathrm{M}$$

Ratio $\dfrac{S_1}{S_2} = \dfrac{6.50 \times 10^{-5}\ \mathrm{M}}{7.07 \times 10^{-7}\ \mathrm{M}} = 91.9.$

Practice Question 12 (Single-Correct MCQ • Selective Sulphide Precipitation Criterion)

The concentration of sulphide ion ($\mathrm{S^{2-}}$) in a $0.1\ \mathrm{M\ HCl}$ solution saturated with $\mathrm{H_2S}$ is $1.0 \times 10^{-19}\ \mathrm{M}$. If $10\ \mathrm{mL}$ of this solution is added to $5\ \mathrm{mL}$ of $0.04\ \mathrm{M}$ separate solutions of $\mathrm{FeSO_4}$, $\mathrm{MnCl_2}$, $\mathrm{ZnCl_2}$, and $\mathrm{CdCl_2}$, in which solutions will precipitation of the metal sulphide occur?
(Given K_{sp} at $298\ \mathrm{K}$: $\mathrm{FeS} = 6.3 \times 10^{-18}$, $\mathrm{MnS} = 2.5 \times 10^{-13}$, $\mathrm{ZnS} = 1.6 \times 10^{-24}$, $\mathrm{CdS} = 8.0 \times 10^{-27}$).

(1) All four solutions ($\mathrm{FeSO_4}$, $\mathrm{MnCl_2}$, $\mathrm{ZnCl_2}$, and $\mathrm{CdCl_2}$)

(2) $\mathrm{FeSO_4}$ and $\mathrm{MnCl_2}$ only

(3) $\mathrm{ZnCl_2}$ and $\mathrm{CdCl_2}$ only

(4) $\mathrm{CdCl_2}$ only

Correct Answer: (3) $\mathrm{ZnCl_2}$ **and** $\mathrm{CdCl_2}$ **only**

Worked Solution: Upon mixing $10\ \mathrm{mL}$ of the $\mathrm{H_2S/HCl}$ solution with $5\ \mathrm{mL}$ of the $0.04\ \mathrm{M}$ metal salt solution, the total volume becomes $15\ \mathrm{mL}$. Immediately after mixing:

$$[\mathrm{M^{2+}}] = 0.04\ \mathrm{M} \times \frac{5\ \mathrm{mL}}{15\ \mathrm{mL}} = 1.33 \times 10^{-2}\ \mathrm{M}, \qquad [\mathrm{S^{2-}}] = 1.0 \times 10^{-19}\ \mathrm{M} \times \frac{10\ \mathrm{mL}}{15\ \mathrm{mL}} = 6.67 \times 10^{-20}\ \mathrm{M}$$

Therefore, the ionic product Q_{sp} in each mixture is:

$$Q_{sp} = [\mathrm{M^{2+}}][\mathrm{S^{2-}}] = (1.33 \times 10^{-2})(6.67 \times 10^{-20}) = 8.87 \times 10^{-22}$$

Precipitation occurs only when $Q_{sp} > K_{sp}$:

- For FeS ($K_{sp} = 6.3 \times 10^{-18}$) and MnS ($K_{sp} = 2.5 \times 10^{-13}$): $Q_{sp} < K_{sp} \rightarrow$ **no precipitation**.
- For ZnS ($K_{sp} = 1.6 \times 10^{-24}$) and CdS ($K_{sp} = 8.0 \times 10^{-27}$): $Q_{sp} > K_{sp} \rightarrow$ **precipitation takes place**.

BOSONANCE

11. Common Pitfalls & High-Yield NEET UG Revision Checklist

Top 6 Exam Pitfalls to Avoid in NEET UG

1. **Counting Solids/Liquids in Δn:** When computing Δn for $K_p = K_c(RT)^{\Delta n}$, count *only gaseous* species! Never include pure solids (s) or pure liquids (l).
2. **Mismatching R and Pressure Units:** Use $R = 0.0831\ \text{bar L mol}^{-1}\text{ K}^{-1}$ when K_p is in bar, $R = 0.0821\ \text{atm L mol}^{-1}\text{ K}^{-1}$ when K_p is in atm, and $R = 8.314\ \text{J mol}^{-1}\text{ K}^{-1}$ when calculating $\Delta G^\circ = -RT\ln K$ in Joules.
3. **Confusing Inert Gas Addition at Constant V vs. Constant P:** Adding an inert gas at *constant volume* never shifts any equilibrium. Only at *constant pressure* (which increases volume) does it shift the reaction toward the side with more moles of gas.
4. **Forgetting Water Auto-Ionization in Dilute Acids/Bases:** For 10^{-7} M or 10^{-8} M strong acid, pH is always slightly less than 7 (e.g., 6.98 for 10^{-8} M HCl), never > 7.
5. **Comparing K_{sp} Directly for Salts of Different Stoichiometry:** You can compare K_{sp} values directly to rank molar solubility *only* if the salts have the exact same formula type (e.g., AgCl vs. AgBr). For $Ni(OH)_2$ ($1:2$, $K_{sp} = 4S^3$) vs. AgCN ($1:1$, $K_{sp} = S^2$), always calculate S explicitly!
6. **Forgetting Volume Dilution When Mixing Solutions:** In precipitation (Q_{sp}) or acid-base neutralization problems, always divide millimoles by the *combined total volume* ($V_1 + V_2$).

Concise Final Formula Recap

- **Equilibrium Constants & Thermodynamics:** $K_p = K_c(RT)^{\Delta n}$; $\quad \Delta G = \Delta G^\circ + RT\ln Q$; $\quad \Delta G^\circ = -2.303\,RT\log K$.
- **Water & Weak Electrolytes (298 K):** $K_w = [\text{H}^+][\text{OH}^-] = 10^{-14}$; $\quad \text{pH} + \text{pOH} = 14$; $\quad K_a \cdot K_b = K_w$; $\quad K_a = \dfrac{c\alpha^2}{1-\alpha}$.
- **Salt Hydrolysis (298 K):** Weak acid + strong base: $\text{pH} = 7 + \frac{1}{2}(\text{p}K_a + \log c)$; $\quad$ Strong acid + weak base: $\text{pH} = 7 - \frac{1}{2}(\text{p}K_b + \log c)$; Weak acid + weak base: $\text{pH} = 7 + \frac{1}{2}(\text{p}K_a - \text{p}K_b)$.
- **Buffers & Solubility:** $\text{pH} = \text{p}K_a + \log\dfrac{[\text{A}^-]}{[\text{HA}]}$; $\quad K_{sp}(\text{M}_x\text{X}_y) = x^x y^y S^{x+y}$; $\quad S(\text{at pH}) = \sqrt{K_{sp}\left(1 + \dfrac{[\text{H}^+]}{K_a}\right)}$.

BOSONANCE